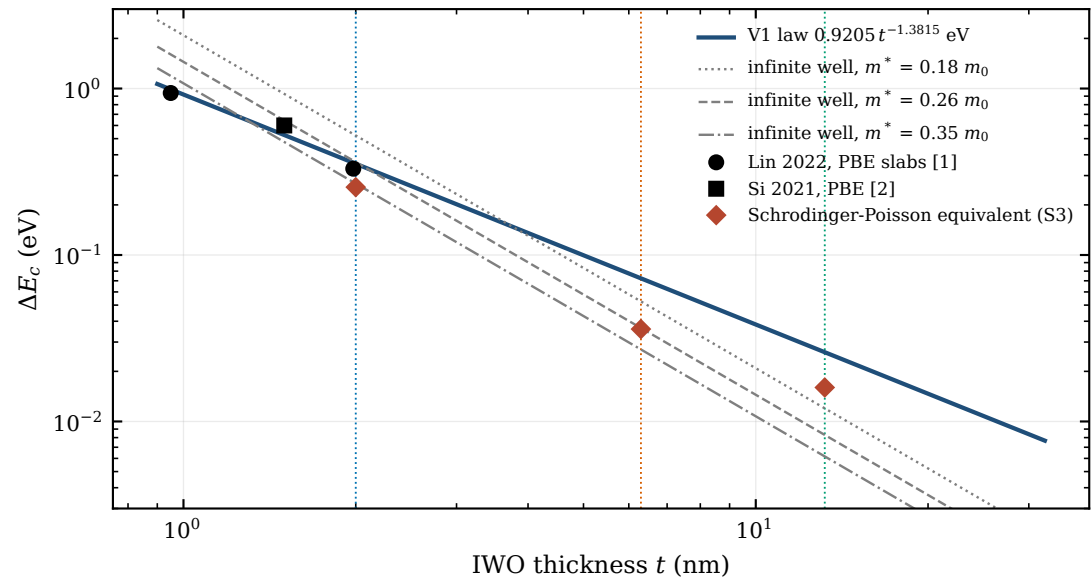


(a) Conduction-band confinement shift $\Delta E_c(t)$, model V1



What is plotted

The upward shift of the conduction-band edge of a thin film relative to bulk. V1 law: log-log fit through the three DFT points. Grey: lowest level of an infinite square well, $E_1 = \hbar^2 \pi^2 / (2m^* t^2)$, the large- t limit; above about 3 nm the V1 power law lies above it. Diamonds: shifts equivalent in subthreshold to a 1-D Schrodinger-Poisson solution (specialist study S3, case Q1, $n_s = 10^{10} \text{ cm}^{-2}$). Dotted lines: the measured films 2.0 / 6.3 / 13.2 nm.

References

[1] Lin et al., ACS Nano 16, 21536 (2022), p. 21542: PBE gap opening +0.94 eV (0.95 nm) and +0.33 eV (1.98 nm). [2] Si et al., Nano Lett. 21, 500 (2021), p. 504: PBE conduction-band shift +0.60 eV for 1.5 nm In_2O_3 on Al_2O_3 . Both are pure In_2O_3 used as a proxy for IWO.